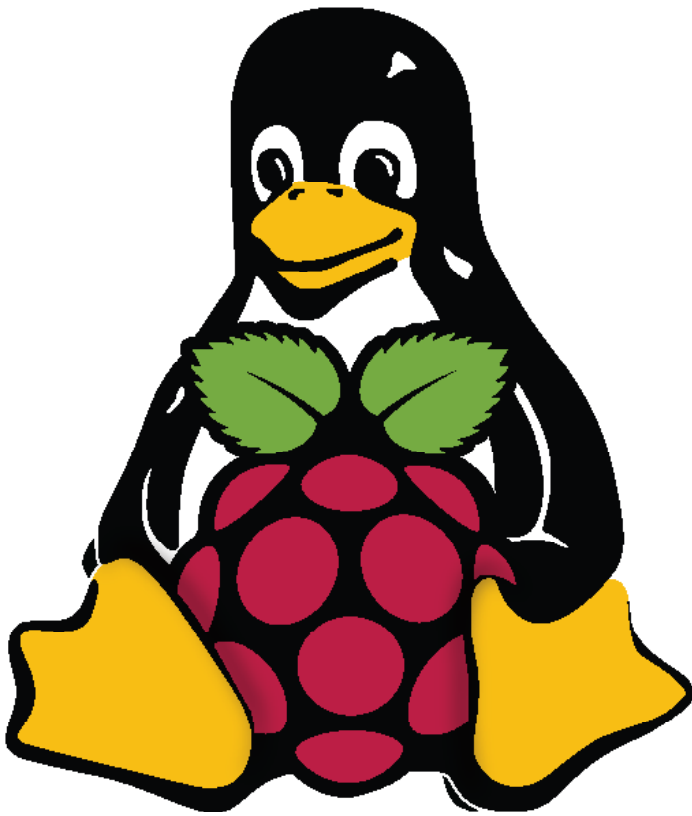


Make:

Linux for

Makers



Understanding the Operating System That Runs
Raspberry Pi and Other Maker SBCs

Aaron Newcomb

Make: Linux for Makers

Some Makers shy away from using the Raspberry Pi or similar boards because they think Linux is too foreign and difficult. The good news is that the Linux operating system is really just another tool in the Maker tool belt! Like all tools, it's not challenging to use once you know how to use it effectively.

It makes sense that Makers would develop a preference for using Linux to help build their projects that run on single board computers, in much the same way we use a screwdriver or hammer in woodworking. In fact, Linux is so powerful, you may start to prefer it to other operating systems and choose to use it on a daily basis!

This is the first book to explain the Linux operating system specifically for Makers, providing a foundation in the basic principles you need for further learning and exploration with your projects.

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- » **Code and write scripts that control real-world hardware**
- » **Learn Linux commands, systems, and processes**
- » **Control GPIO pins on your Raspberry Pi**
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- » **Run a virtual Raspberry Pi on Windows, Mac, or Linux**

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Linux for Makers

**Understanding the
Operating System That
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Other Maker SBCs**

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by Aaron Newcomb

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Contents

Preface.	ix
1/Getting Started.	1
Choosing and Downloading a Disk Image.	2
Uncompressing the Disk Image.	3
Windows.	3
MacOS.	4
Linux.	5
Writing the Disk Image to the SD Card.	7
Windows.	7
MacOS.	8
Linux.	11
Booting the Raspberry Pi for the First Time.	13
Expanding the Filesystem.	14
Changing the Localization Options.	15
Changing the Default Password.	19
Why This Matters for Makers.	20
2/Linux Principles.	21
The Linux Desktop.	21
The Terminal or Console.	23
The Shell in a Nutshell.	25
Try It for Yourself.	26
Filesystems and Structures.	28
Try It for Yourself.	28
Users and Groups.	31
Permissions and sudo.	33
Try It for Yourself.	35
Try It for Yourself.	36
Services.	36
Try It for Yourself.	37
Processes.	38
Try It for Yourself.	39

Why This Matters for Makers.....	40
3/Using the Desktop.....	41
When to Use the Desktop?.....	41
When Not to Use the Desktop?.....	42
Understanding the Layout.....	43
Connecting to the Network.....	44
Changing the Look and Feel.....	45
Changing the Panel Location.....	45
Changing the Background Image.....	45
Changing the Shortcuts in the Application Launch Bar.....	46
Creating a Desktop Shortcut.....	47
Why This Matters for Makers.....	49
4/Command-Line Basics.....	51
Understanding the Prompt.....	51
Try It for Yourself.....	52
Orienting Yourself in the Filesystem.....	53
Where Am I?: pwd.....	53
Changing the Working Directory: cd.....	54
Printing Out the Contents of a Directory: ls.....	56
Creating New Files and Directories: mkdir and touch.....	58
Moving and Deleting Files: cp, mv, and rm.....	59
Try It for Yourself.....	60
Get Help with a Command: help, man, and info.....	62
Try It for Yourself.....	69
Eliminate Some Typing.....	69
Auto-Complete a Command: Tab.....	70
Search for a Previous Command: Up, Ctrl-R.....	71
Try It for Yourself.....	72
Connecting to the Network via the Command Line.....	73
The Interfaces File.....	73
Wired Ethernet.....	75
Static IP Address.....	75
WiFi.....	76
More Secure WiFi with Multiple Networks.....	76
Installing Software: apt.....	79
Using apt-get update.....	80
Using apt-get upgrade.....	81
Using apt-cache.....	85
Using apt-get install.....	86
apt-get remove.....	87

apt-get dist-upgrade.	88
Fixing Conflicts.	89
Try It for Yourself.	89
Rebooting and Shutting Down.	90
Why This Matters for Makers.	91
5/Headless Operation.	93
Turning Off the Desktop.	93
Finding Your System on the Network.	95
Raspberry Pi.	96
Router.	97
Android/iPhone.	98
Command-Line Access: ssh.	99
Windows.	101
MacOS.	104
Linux.	106
Android/iPhone.	106
Remote Desktops: vnc.	108
Setting Up the Raspberry Pi.	108
Windows.	110
MacOS.	112
Linux.	114
Android/iPhone.	116
Transferring Files: scp, sftp.	117
Windows.	118
MacOS.	119
Linux.	122
From the Command Line: MacOS and Linux.	125
Why This Matters for Makers.	126
6/Tips and Tricks.	127
Changing Your Hostname.	127
Starting a Script on Bootup: rc.local.	131
Try It for Yourself.	132
Aliases.	133
Try It for Yourself.	135
Checking Disk and File Space Usage: df, du.	136
Performance Monitoring: top.	138
Try It for Yourself.	145
Killing a Process: Ctrl-C, ps, kill.	145
Stop, Background, and Foreground Jobs: Ctrl-Z, &, fg.	148
Try It for Yourself.	150

Finding USB Devices: lsusb.	151
Logging the Output of a Script: >, >>.	153
Searching the Output of a Command: grep.	155
Monitoring a Log File: tail.	157
Adding a User: adduser, addgroup.	158
Changing File Ownership and Permissions: chown, chmod.	159
Running More Than One Command at the Same Time: &&, 	162
Opening Another Console Session.	163
Dealing with Long Commands.	164
Scheduling Jobs: cron.	165
Why This Matters for Makers.	168
7/Controlling the Physical World.	169
GPIO.	169
I ² C and SPI.	177
Talking to Arduino.	183
Why This Matters for Makers.	187
8/Using Multimedia.	189
Choosing HDMI or Analog.	189
Playing Audio and Video Files.	191
Controlling the Volume.	192
Playing Media from a Script.	193
Why This Matters for Makers.	194
9/Accessing Cloud Services.	195
Cloud Storage Services from the Command Line.	195
IFTTT.	199
Try It for Yourself.	208
Run a Dedicated Web Server.	208
Installation.	209
Configuration for Python.	209
Test It Out.	209
Roll Your Own.	211
Nimbus.	211
Tonido.	213
Why This Matters for Makers.	216
10/Virtual Raspberry Pi.	217
Requirements.	218
Installation.	218

Usage.....	219
Why This Matters for Makers.....	221
A/Linux Background.	223
Index.	239

Preface

When I started a Makerspace in my local community, I noticed some particularly interesting learning trends. Some people were reluctant to learn a new skill until someone shared some basic techniques that helped get them started down the path of understanding. Other users would jump right into learning a new skill without any idea of what they were doing. This would lead to slow progress until, again, someone provided some assistance that would lead them in the right direction. In both cases, just a little guidance in the beginning greatly accelerated the learning process.

Learning how to use Linux for making and building projects is no easy task. In many cases, Makers cut and paste from a website tutorial into the Linux command line without understanding what they are actually doing, only to be frustrated when they want to modify or tweak something to suit their needs. Also, many Makers shy away from using the Raspberry Pi or similar boards because they feel Linux is too foreign and that using a command line as indicated in many tutorials will be more difficult than using a GUI.

This book aims to overcome those fears and provide a foundation for further learning and exploration when using the Linux operating system for your projects. Linux is just another tool in your Maker tool belt. It might be different from other operating systems you've used in the past, but—like all tools—it's no more challenging to use once you know how to use it effectively. In fact, Linux is so powerful, you may start to prefer it to other operating systems and choose to use it on a daily basis.

Linux is a powerful open source operating system that has been around for many years and is widely used for running servers and websites. But most students and Makers encounter it for the first time when they're working on projects with their Raspberry Pi or similar single-board computer (SBC), such as

BeagleBone Black or Intel Galileo. *Linux for Makers* is the first book that explains the Linux operating system specifically for Makers as opposed to programmers and administrators. By gaining a deeper understanding of Linux, Makers can add another useful tool to their kit that will help them build projects more easily.

Because this book was written with today's Maker in mind, it will focus mostly on the Raspbian distribution of Linux running on the Raspberry Pi, as that platform is the most prolific in the ecosystem today. However, most of the topics covered will apply broadly to other Linux distributions, and I will indicate when they may differ. To that end, this book will focus on the basic principles that a Maker needs to know, avoiding details that are not particularly relevant to building projects. After loading the operating system, I will cover the principles of how Linux works, how to use the command line, how to control devices, and loads of tips and tricks that can help you be more effective.

Throughout the book, you will find sections called “Try It for Yourself” where you can get your hands dirty practicing what you just learned and explore additional opportunities to try out new concepts. I have also included illustrations and pictures throughout the book that should help clarify what you can expect to see as you use Linux on your Raspberry Pi.

I have also included a brief history of Linux in [Appendix A](#) for those readers who might be wondering “How did all this get started?” or “How did Linux end up getting put together the way it did?”

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

Constant width

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function

names, databases, data types, environment variables, statements, and keywords.

Constant width bold

Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows text that should be replaced with user-supplied values or by values determined by context.



This element signifies a tip, suggestion, or a general note.



This element indicates a warning or caution.

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Make: unites, inspires, informs, and entertains a growing community of resourceful people who undertake amazing projects in their backyards, basements, and garages. Make: celebrates your right to tweak, hack, and bend any technology to your will. The Make: audience continues to be a growing culture and community that believes in bettering ourselves, our environment, our educational system—our entire world. This is much more than an audience; it's a worldwide movement that Make is leading. We call it the Maker Movement.

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Thanks to James who introduced me to Linux back in 1997. Know someone who might like Linux or Raspberry Pi? Tell them about it!

I am grateful to the support of my editor, Patrick, and all the staff at Maker Media and O'Reilly Media who guided me through the writing, editing, and reviewing process.

I also want to add a big shoutout to the people who gave up their time to help review the book and offer so many great suggestions—Robert Shaver, Christoph Zimmermann, Jim Kennon, Rashed Harun, and Broedy Bowers.

1/Getting Started

The Raspberry Pi is a single-board computer (SBC), which means that—as the name suggests—it is a complete computer system built on a single printed circuit board (PCB). Like most SBCs, it doesn't come out of the box ready to power up and use. It has the same basic components built into the board as any other computer has: a central processing unit (CPU), memory, video processor, audio, and networking.

The one thing it doesn't have out of the box is a storage device. On a computer or laptop, most people use a hard drive that contains the operating system and all their files. With a Raspberry Pi, you use an SD card as the main storage device. So before we can tackle the ins and outs of making things using Linux, we need to load the operating system you want to use with the Raspberry Pi on the SD card. For best results, you should use an SD card that has at least 8 GB of available capacity.

What Is a Disk Image?

A *disk image* is a single file that represents a point-in-time copy of an entire storage device. Just like a photograph is an image that can contain different people or objects, a disk image can contain lots of different partitions, directories, and files.

This process can be confusing for some people, so let's break it down into individual steps and take them one at a time. These steps include downloading a compressed disk image from the internet, uncompressing that image on your local computer, writing that image to your SD card, and finally, booting up your Raspberry Pi. You will find that these steps apply to other SBCs as well, although the exact image file you use will change because the operating system needs to be built specifically for the board you're using.

I will be mentioning some concepts in this chapter that may be new to you, like filesystems, terminal emulators, and the command line. Don't worry. We will cover these in depth in [Chapter 2](#). For now, we just need to get things running so you can use your Raspberry Pi as you go through this book. In order to get started, you will need access to a desktop or laptop computer and a connection to the internet.

Choosing and Downloading a Disk Image

The best place to find the most up-to-date disk images for the Raspberry Pi is on the Raspberry Pi Foundation website at <http://raspberrypi.org/downloads>. When you get to that site, you will see that there are a number of disk images to choose from. The two that have official support from the Raspberry Pi Foundation are NOOBS and Raspbian. *NOOBS* is essentially an installer for Raspbian as well as several other operating systems that can be run on the Raspberry Pi. *NOOBS* automates several of the tasks described in this chapter, but is not the actual operating system you will end up using. Instead, the operating system is downloaded as part of the installation process. *Raspbian*, on the other hand, contains the actual operating system, and the installation method requires you to manually write the disk image to an SD card.

I recommend choosing the Raspbian image for a number of reasons. First, while installing *NOOBS* might seem less complicated, once you learn the standard installation procedure, Raspbian will actually be easier and faster than using *NOOBS*. Second, learning the installation process for Raspbian will teach you how to load any disk image you might want to try in the future, and learning how to do things is what this book is all about. Third, since loading Raspbian using this process results in a more standard disk layout and structure on the SD card, it makes it a little easier to back up your image for safekeeping. This is something you definitely want to do to protect your work and prevent data loss in case you break your SD card or damage your Raspberry Pi.

Click on the Raspbian link and download the Raspbian Jessie file. Later, if you decide you don't want to use the desktop (see [Chapter 3](#)) you can choose the Raspbian Jessie Lite file, which is smaller in size but does not come prepackaged with a desktop environment. Save the ZIP file in a place you can find easily later, like the *Downloads* folder. At the time of this writing the filename looks like this: *2017-01-11-raspbian-jessie.zip*.

Uncompressing the Disk Image

Disk images are compressed to make them smaller for downloading. Compressing a file basically uses an algorithm to take out all the empty spaces and duplicate existing information. For example, imagine removing all the spaces between the words of an essay or a text to a friend. It makes the finished product much smaller, but also much more difficult to read. So in order to make use of the disk image, you will first need to uncompress it. You will need software in order to uncompress the disk image, and in most cases that software is already available on your desktop or laptop.

Windows

In recent versions of Windows, the decompression software you need is already built into File Explorer. Simply open it up and navigate to the folder where you saved the ZIP file. Double-click on the ZIP file to open it. Clicking on the Extract button will save the disk image to a place of your choosing that you can easily find later. This could be the same folder where you saved the ZIP file originally (see [Figure 1-1](#) for an example of what this looks like).

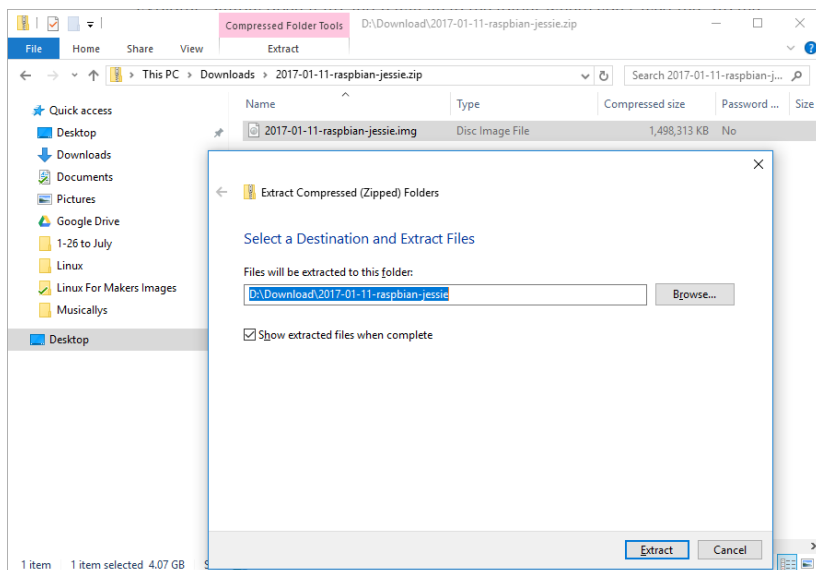


Figure 1-1. *Extracting the disk image using Windows*

MacOS

When you download the ZIP file with a recent version of Safari, the ZIP file will be automatically uncompressed and saved in your *Downloads* folder by default. If you are using an older version, macOS has a built-in tool for uncompressing files called Archive Utility. Just double-click on the ZIP file you downloaded, and Archive Utility will uncompress the file for you (see [Figure 1-2](#)).

When the process is complete, you will see the extracted *.img* file appear on your desktop or wherever you saved your ZIP file.

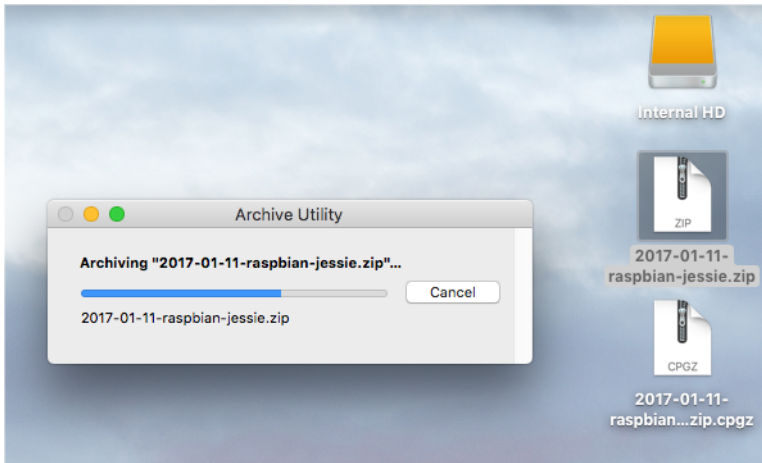


Figure 1-2. *The Archive Utility running on macOS*

Linux

Most distributions of Linux also come with built-in programs to extract compressed files. On the desktop, you can open the file browser and double-click on the ZIP file you downloaded. This will open up Archive Manager and allow you to extract the image file (see [Figure 1-3](#)).

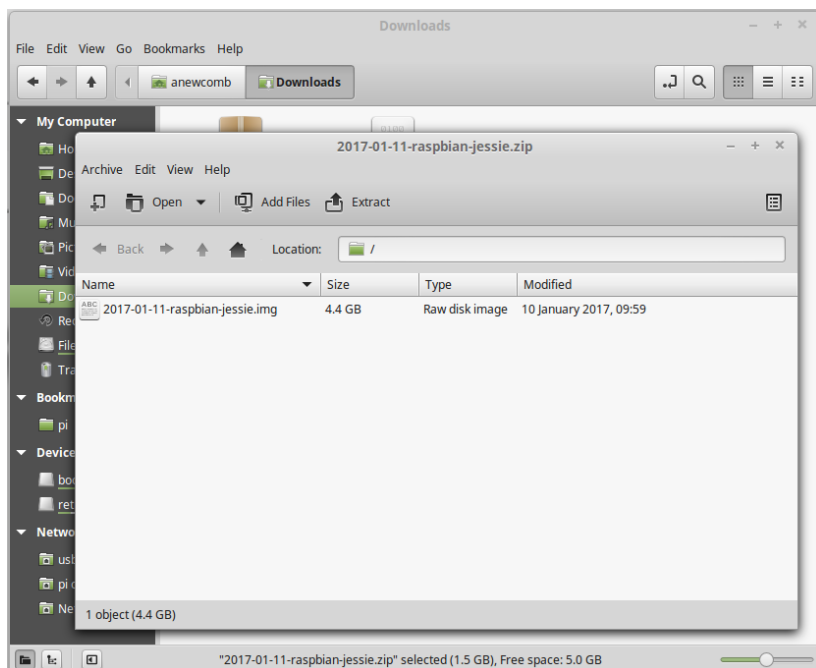


Figure 1-3. *Extracting the disk image using Linux*

You can also do this just as easily from the command line. Open a terminal emulator and type:

```
cd Downloads
unzip 2017-01-11-raspbian-jessie.zip
```

This assumes that you saved the ZIP file to your default download location and that the filename is correct. Be patient. This is a big file and uncompressing it will take some time. See [Figure 1-4](#) to see what this looks like.

Command-Line Confusion

Not comfortable using the command line yet? That's a big part of what this book is about. You will learn more about that in the next few chapters.

```
anewcomb@anewcomb-VirtualBox ~ $ cd Downloads
anewcomb@anewcomb-VirtualBox ~/Downloads $ unzip 2017-01-11-raspbian-jessie.zip
Archive: 2017-01-11-raspbian-jessie.zip
  inflating: 2017-01-11-raspbian-jessie.img
anewcomb@anewcomb-VirtualBox ~/Downloads $
```

Figure 1-4. *Extracting the disk image using the Linux command line*

Writing the Disk Image to the SD Card

Be aware that after this step, any files you may have had on your SD card will be deleted.

Windows

Windows is currently the easiest operating system to use when writing a disk image to a card. However, unlike macOS and Linux, it requires you to download some software first. Open your browser and download the [Win32 Disk Imager application](#).

Install the application by double-clicking on the file you downloaded. After it installs successfully, connect your SD card to your computer and make note of the drive letter that Windows assigns to it. Now, open the application. The first thing to check is that the drive letter the application selects is actually the drive letter that corresponds to your SD card. Win32 Disk Imager is pretty good about only selecting SD cards, but always double-check this because you are about to erase the drive, and you definitely don't want to erase your C: drive on Windows.

Now click on the blue folder icon to select the image file you extracted in the previous step. See [Figure 1-5](#) for an example of what this should look like.

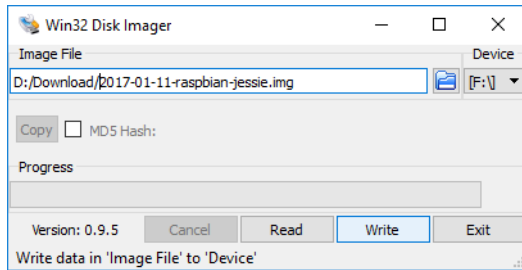


Figure 1-5. *The Win32 Disk Imager program interface*

When you've selected the correct file, you can click the Write button. This will overwrite all the data on the SD card. When the process is completed, close the Win32 Disk Imager application. Open File Explorer, right-click on the SD card drive letter, and select Eject. Always do this before removing your SD card to make sure the computer is done writing files in the background.

MacOS

Like Linux, macOS already has all the software you need to write the image to an SD card. However, you will need to use the command line. Open Finder and select Applications→Utilities→Terminal (see [Figure 1-6](#)).

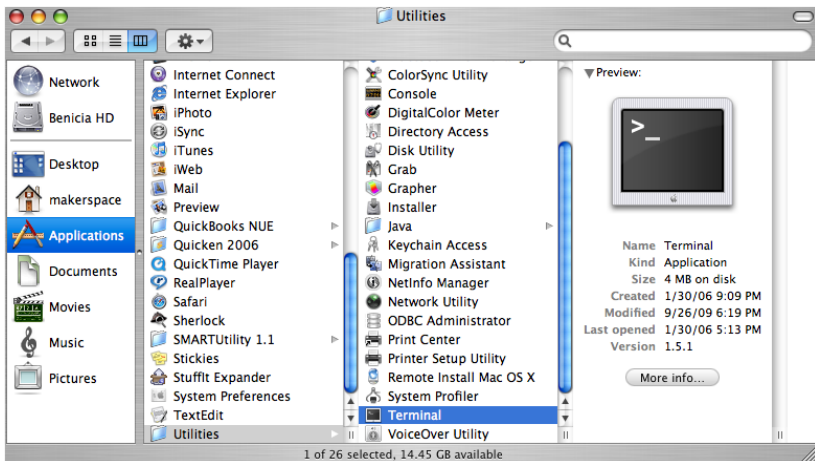
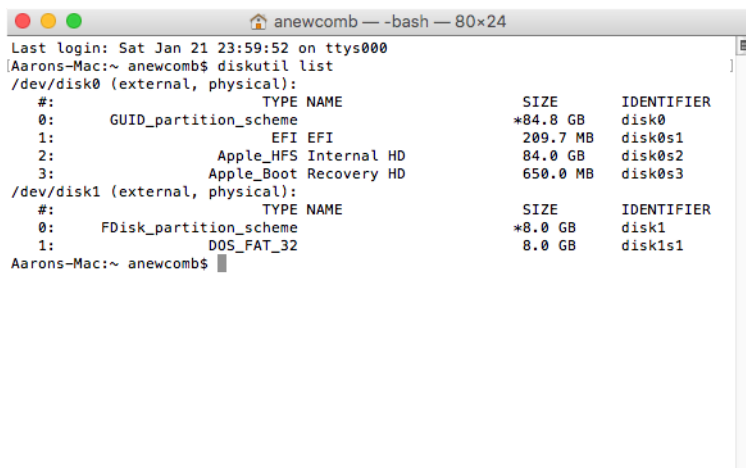


Figure 1-6. *Finding the Terminal program on macOS*

Insert the SD card and wait a few moments for macOS to recognize it. Use the `diskutil list` command in the terminal window to print a list of all the disks attached to your Mac:

```
diskutil list
```

Identify the disk (not partition) representing your SD card (e.g., `disk1`, not `disk1s1`) as shown in [Figure 1-7](#).



```
anewcomb — -bash — 80x24
Last login: Sat Jan 21 23:59:52 on ttys000
[Aarons-Mac:~ anewcomb$ diskutil list
/dev/disk0 (external, physical):
#    TYPE NAME              SIZE      IDENTIFIER
0:    GUID_partition_scheme  *84.8 GB  disk0
1:            EFI EFI                209.7 MB  disk0s1
2:            Apple_HFS Internal HD    84.0 GB  disk0s2
3:            Apple_Boot Recovery HD    650.0 MB  disk0s3
/dev/disk1 (external, physical):
#    TYPE NAME              SIZE      IDENTIFIER
0:    FDisk_partition_scheme  *8.0 GB   disk1
1:            DOS_FAT_32                8.0 GB   disk1s1
Aarons-Mac:~ anewcomb$
```

Figure 1-7. Example output of the `diskutil list` command

In this case, I have a 64 GB SD card and macOS recognizes it as `disk1`.

Unmount your SD card by using the `diskutil unmountDisk` command and the disk name to prepare for copying data to it (see [Figure 1-8](#):

```
diskutil unmountDisk /dev/disk1
```

```
aneucomb — -bash — 80x24
Last login: Sat Jan 21 23:59:52 on ttys000
Aarons-Mac:~ anewcomb$ diskutil list
/dev/disk0 (external, physical):
#:

| #: | TYPE                  | NAME        | SIZE     | IDENTIFIER |
|----|-----------------------|-------------|----------|------------|
| 0: | GUID_partition_scheme |             | *84.8 GB | disk0      |
| 1: | EFI                   | EFI         | 209.7 MB | disk0s1    |
| 2: | Apple_HFS             | Internal HD | 84.0 GB  | disk0s2    |
| 3: | Apple_Boot            | Recovery HD | 650.0 MB | disk0s3    |


/dev/disk1 (external, physical):
#:

| #: | TYPE                   | NAME | SIZE    | IDENTIFIER |
|----|------------------------|------|---------|------------|
| 0: | FDisk_partition_scheme |      | *8.0 GB | disk1      |
| 1: | DOS_FAT_32             |      | 8.0 GB  | disk1s1    |


Aarons-Mac:~ anewcomb$ diskutil unmountDisk /dev/disk1
Unmount of all volumes on disk1 was successful
Aarons-Mac:~ anewcomb$
```

Figure 1-8. Using *diskutil* to unmount a disk

Now we need to copy the image file over to the SD card. We will use the *data duplicator*, or *dd* command, for this. Be very careful to get the right disk number so that you don't overwrite your system disk! You will also need to use *SuperUserDO* (*sudo*) for this command. *sudo* is a secure way for a regular user to run a command that normally requires administrator privileges:

```
sudo dd if=Desktop/2017-01-11-raspbian-jessie.img of=/dev/rdisk1 bs=1m
```

You will be asked for your password, since you are using *sudo*. This may result in an error if you have GNU coreutils installed:

```
dd: invalid number '1m'
```

Don't worry about what that means just yet—all you need to do is use a block size of *1m* in the *bs=* section of the command, as follows:

```
sudo dd if=Desktop/2017-01-11-raspbian-jessie.img of=/dev/rdisk1 bs=1m
```

This will take quite a few minutes, depending on the image file size. You can check the progress by pressing *Ctrl-T* to send a *SIGINFO* signal (see [Figure 1-9](#)). If this command still fails, try using the *disk* command instead of *rdisk*.

```
aneucomb — dd * sudo — 80x29
Last login: Sat Jan 21 23:59:52 on ttys000
Aarons-Mac:~ anewcomb$ diskutil list
/dev/disk0 (external, physical):
#:                       TYPE NAME              SIZE       IDENTIFIER
0:      GUID_partition_scheme             *84.8 GB    disk0
1:                  EFI EFI                 209.7 MB    disk0s1
2:              Apple_HFS Internal HD         84.0 GB    disk0s2
3:              Apple_Boot Recovery HD         650.0 MB    disk0s3
/dev/disk1 (external, physical):
#:                       TYPE NAME              SIZE       IDENTIFIER
0:      FDisk_partition_scheme             *8.0 GB     disk1
1:                  DOS_FAT_32                8.0 GB     disk1s1
Aarons-Mac:~ anewcomb$ diskutil unmountDisk /dev/disk1
Unmount of all volumes on disk1 was successful
Aarons-Mac:~ anewcomb$ sudo dd if=Downloads/2017-01-11-raspbian-jessie-lite.img
of=/dev/rdisk1 bs=1m

WARNING: Improper use of the sudo command could lead to data loss
or the deletion of important system files. Please double-check your
typing when using sudo. Type "man sudo" for more information.

To proceed, enter your password, or type Ctrl-C to abort.

Password:
load: 1.37 cmd: dd 538 uninterruptible 0.00u 0.60s
549+0 records in
548+0 records out
574619648 bytes transferred in 71.079200 secs (8084217 bytes/sec)
```

Figure 1-9. Using the *dd* utility on macOS

In this case, the *dd* command has transferred 549 1 MB blocks. Once the process is complete, the terminal will bring you back to a prompt. Finally, run one last command before you disconnect your SD card:

```
sync
```

This will make sure all the writes to the SD card that may be occurring in the background have finished. You can now remove your SD card.

Linux

Before you connect your SD card to your computer, run this command:

```
sudo fdisk -l
```

This will show you all the physical drives connected to your system. Note the size of the drive and the name of the disk. See [Figure 1-10](#).